

Beyond the Speech: Context-Aware Topic Labelling and Linguistic Style in Parliamentary Debates

Pavle Savković¹ Stefan Thalmann² Armin Spök³ Denis Helic^{1,*}

¹ Institute of Human-Centred Computing, Graz University of Technology, Graz, Austria

² Center for Business Analytics and Data Science, Department of Operations and
Information Systems, University of Graz, Graz, Austria

³ Science, Technology and Society Unit, Institute of Human-Centred Computing, Graz
University of Technology, Graz, Austria

*Corresponding author: dhelic@tugraz.at

Author	ORCID	Google Scholar
Pavle Savković	0009-0007-8214-2369	Profile
Stefan Thalmann	0000-0001-6529-7958	Profile
Armin Spök	0000-0003-2039-3553	Profile
Denis Helic	0000-0003-0725-7450	Profile

Abstract

Parliamentary debates are not a sequence of independent speeches; they are structured, turn-taking interactions whose meaning depends on procedural context. Yet most large-scale political text analysis still treats speeches as isolated documents. We propose a context-aware, segment-first pipeline that groups speeches into coherent agenda episodes and assigns policy topics at the episode level. Using PARLAMINT data from Austria, Great Britain, and Croatia (1,407,009 speeches), we combine procedural cues and semantic similarity to segment sittings, cluster episode embeddings, and map the resulting clusters to Comparative Agendas Project (CAP) domains. Linguistic style is then profiled with LIWC-22. The episode-level labels yield substantively interpretable results: macroeconomic debate is consistently money-heavy and less moralized; institutional role shapes style more strongly than ideology or demographics, a surprising finding given the expectation that ideology should leave a strong rhetorical footprint; and major crises such as COVID-19 and the Ukraine war temporarily reorganize both agenda attention and rhetoric before both return toward the institutional baseline.

Keywords: parliamentary discourse, political language, topic labelling, agenda episodes, linguistic style, LIWC

1 Introduction

Parliamentary debates are dialogues; what a person says is shaped by the speaker before them, and it shapes the speaker after them. Debate is a procedural, turn-taking interaction in which speakers respond to one another under an agenda set by the chamber. What a short intervention means often depends on the surrounding exchange, not just on the words in that turn alone [1–3]. Yet much computational work on political text still treats speeches as self-contained documents and assigns labels one speech at a time [4–6].

That mismatch matters. In parliamentary data, many interventions are brief, procedural, reactive, or narrowly framed. Read in isolation, they can look ambiguous or even off-topic. Read in context, they are often obvious parts of the same agenda item. A speech-level classifier therefore tends to create label switches inside what participants would recognize as one continuous debate. This hurts interpretability at exactly the point where agenda reconstruction and topic-conditioned style analysis need stability most.

Table 1 shows the problem in concrete form. Three consecutive speeches in the British parliament all belong to the same Energy Bill discussion. A minister announces the bill, another politician raises concerns about fuel poverty and house heating, and another asks about environmental concerns regarding fracking. To anyone following the debate, these are parts of one Energy episode. But a speech-level classifier, reading each turn separately, assigns three different labels: Energy, Social Welfare, and Environment.

Table 1: Three consecutive speeches from a Great Britain Energy Bill debate. A speech-level classifier assigns a different label to each turn read in isolation. An episode-level approach recognizes the ongoing Energy debate and assigns a consistent label to all three.

Speaker	Text	PARLACAP topic	Our topic
Rudd, Amber	Our Energy Bill receives Royal Assent today. It is a vital part of our plan to ensure that families and businesses have access to secure, clean and affordable energy. We are delivering on our commitment to end subsidies for onshore wind and supporting the Oil and Gas Authority with powers to drive collaboration.	<i>Energy</i>	<i>Energy</i>
Abrahams, Deborah	Evidence shows that sick and disabled people are at particular risk of fuel poverty, especially after the recent social-security cuts. Will the Secretary of State look at better targeting of warm home discount funding?	<i>Social Welfare</i>	<i>Energy</i>
Tomlinson-Mynors, M.	I have been contacted by constituents concerned about fracking in Dorset. What reassurance can the Minister give about environmental considerations, public consultation, and ensuring fracking is considered only in appropriate locations?	<i>Environment</i>	<i>Energy</i>

Our claim is simple: for this kind of data, the more appropriate unit is not the isolated speech but the coherent *agenda episode*. This paper follows the logic of that example. We first recover coherent agenda episodes, then assign policy topics to those episodes, and only then analyze linguistic style.

The study addresses three research questions:

1. How does policy topic shape linguistic style in parliamentary debate?
2. Do institutional role and demographic characteristics produce systematic rhetorical differences?
3. How do major external shocks affect parliamentary agenda attention and linguistic style over time?

Using 1,407,009 speeches from Austria, Great Britain, and Croatia, we segment sittings into agenda episodes, embed and cluster those episodes, map clusters to CAP domains,

and profile linguistic style with LIWC-22 (Linguistic Inquiry and Word Count). The paper makes three contributions. First, it introduces a sequence-aware topic-labelling pipeline built around parliamentary discourse structure rather than isolated-speech classification. Second, it argues that speech-level benchmark scores are useful but incomplete for evaluating an episode-level method. Third, it shows that the resulting labels support interpretable findings about topic-specific rhetorical styles, institutional role differences, and crisis-driven shifts in attention and rhetoric.

1.1 Related work

Research on parliamentary discourse has long stressed that debate is shaped by institutional procedure, turn-taking, and agenda structure [1–3, 7]. The implication is theoretical as well as descriptive: if meaning in parliament is produced through sequential interaction under procedural rules, treating each speech as a self-contained document strips away part of what makes parliamentary language political in the first place. Computational work still defaults to the speech as the basic unit for tasks such as sentiment, position-taking, and stance [8–12], a unit that is reasonable for many questions but less suited to recovering parliament’s ongoing agenda process.

Our approach connects two areas of research. The first is parliamentary agenda coding, enabled by resources such as PARLAMINT and PARLACAP, which provide comparable corpora and CAP-style policy labels across countries [13–16]. The second is text segmentation, where topic boundaries are inferred from cohesion shifts or semantic drift [17–21]. We bring these together by adapting segmentation to parliamentary procedure through a hybrid of chair cues and semantic transitions, treating segmentation not only as a technical step but as a way of reconstructing the agenda-organized interactional units through which parliamentary politics is actually conducted.

Prior work has also used LIWC-22 and related tools to study affect, morality, complexity, and authority in political language [22–27]. What is often missing is a stable topical frame. Our workflow treats topic labelling and style profiling as one problem: recovering coherent episodes makes it possible to ask not only *what* parliament talks about, but *how* institutional position, policy domain, and crisis conditions shape the language through which politics is performed.

2 Methodology

2.1 Data

The analysis uses PARLAMINT proceedings from Austria, Great Britain, and Croatia [14]. These cases vary in language, institutional practice, and political context while remaining

comparable through a shared corpus framework. Table 2 summarizes the material. We distinguish between *speeches*, the transcript units supplied by the corpus, and *agenda episodes*, analytically reconstructed stretches of discussion on the same policy issue. The three-country design is deliberate: finding the same patterns across languages and institutional settings is substantially stronger evidence of robustness than a single-country result.

Table 2: Descriptive statistics of the analyzed corpora.

Country	Period	Sitting days	Speeches
Austria (AT)	1996–2022	1,221	231,759
Croatia (HR)	2003–2022	1,708	504,338
Great Britain (GB)	2015–2022	2,209	670,912
Total		4,627	1,407,009

We evaluate against two reference sources: full-corpus PARLACAP labels and balanced human-labelled test sets for Great Britain and Croatia [15]. The human test sets contain 40 instances per CAP code, exclude chairperson speeches, and annotate speeches in isolation – a detail central to the evaluation logic in Section 2.5.

2.2 Segmentation into agenda episodes

Each sitting is segmented with a hybrid boundary detector designed for the way parliamentary debate actually unfolds. Some topic changes are explicit: the chairperson formally closes one agenda item and announces the next. Others are gradual: one line of discussion fades while another emerges, with no single turn that cleanly marks the transition. Our procedure is built to capture both cases.

The first signal is procedural. We detect candidate boundaries from chairperson interventions that contain language-specific agenda markers, such as phrases that announce a move to the next item or reopen a different item on the floor. These markers are treated as strong boundary candidates because they reflect the chamber’s own procedural organization. The second signal is semantic. For each candidate position i in a sitting, we compare the mean embedding of the k speeches before i with the mean embedding of the k speeches after i , and compute the cosine similarity between the two window representations. A large drop in similarity indicates that the discussion on the left and right of that position has become substantively different. Intuitively, if speeches $S_1 \dots S_5$ concern an Energy Bill and speeches $S_6 \dots S_9$ shift to environmental regulation, the similarity across the S_5/S_6 boundary should be lower than the similarity within either block.

Rather than fixing the semantic window size globally, we optimize it separately for each country. On a held-out sample of 20% of sittings (with a minimum of 10 and a

maximum of 50 sampled sittings), we test window sizes from 1 to 10 speeches. For each candidate value of k , we first detect semantic boundaries at positions whose similarity drop falls in the top 5% of that sitting’s drop distribution. We then combine semantic and procedural cues conservatively: a chair-based boundary is kept only if a semantic boundary occurs within ± 3 speeches, while semantic boundaries are retained directly. This favors precision over aggressive splitting and prevents isolated chair interventions from creating spurious breaks.

To choose the best window size, we score each candidate segmentation on the held-out sample using a composite criterion. One part rewards agreement with chair-indicated agenda transitions; the other rewards semantic quality, defined by high within-segment coherence and high separation from the following segment in embedding space. We weight the procedural and semantic components equally and select the window size with the highest combined score. Sittings with fewer than five speeches are left unsplit, since there is too little material for reliable internal boundary detection. The output is a sequence of coherent agenda episodes, which then becomes the unit for clustering, CAP mapping, and temporal aggregation.

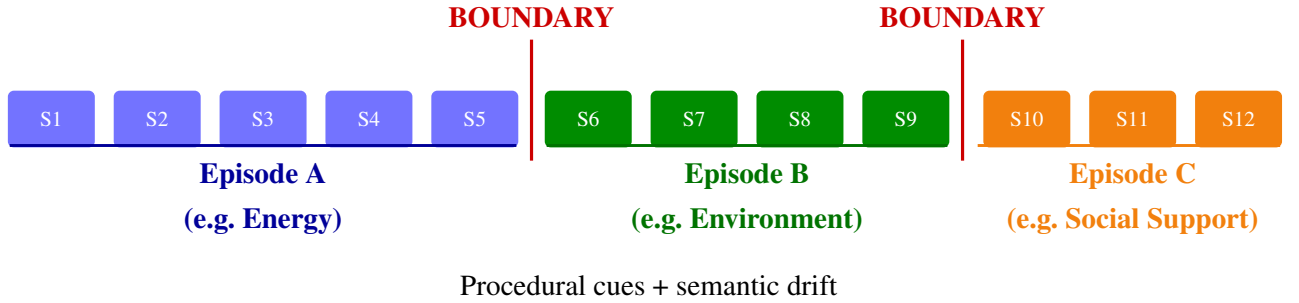


Figure 1: **Illustration of agenda segmentation within a sitting.** Sequences of speeches are grouped into coherent agenda episodes. Boundaries can be identified either by explicit procedural cues, such as chairperson agenda announcements, or by sharp semantic shifts between neighbouring windows of speeches.

2.3 Embeddings, clustering, and CAP alignment

Texts are embedded with the multilingual BAAI/bge-m3 model, which produces 1024-dimensional vectors and supports up to 8,192 tokens [28]. Longer texts are chunked with overlap and averaged, which matters mainly at the segment level rather than for individual speeches.

Episode embeddings are then reduced with UMAP to 10 dimensions and clustered with a Gaussian Mixture Model, which allows soft assignments and non-spherical cluster structure [29, 30]. The number of components is selected by silhouette score, yielding 180 clusters for Austria, 170 for Great Britain, and 185 for Croatia.

For each cluster, we derive a keyword representation using class-based TF-IDF and

extract the top 15 unigrams and bigrams [31]. These keyword lists are then passed to an LLM-based mapping step that returns both (i) a short English topic name and (ii) the single best-fitting CAP major-domain label. The prompt provides the country, source language, cluster keywords, and the full list of available CAP categories, and instructs the model to respond in a fixed two-line format; the full template is reproduced in Appendix A. We use this step because manually mapping 170–185 clusters per country is labor-intensive, while the input to the model is already highly compressed and interpretable: the top terms of each cluster. In a 30-cluster spot check, the model’s CAP assignments agreed with our own independent judgments in all 30 cases on the same keyword lists. For example, a cluster with top terms such as *hospital*, *vaccine*, *patient*, and *doctor* is assigned **Health**; one dominated by *budget*, *deficit*, and *fiscal* is assigned **Macroeconomics**. The remainder of the pipeline is fully deterministic – fixed inputs produce fixed embeddings, clusters, and keywords – and the keyword list, prompt template, and model output are logged for every cluster, making the final label assignment auditable.

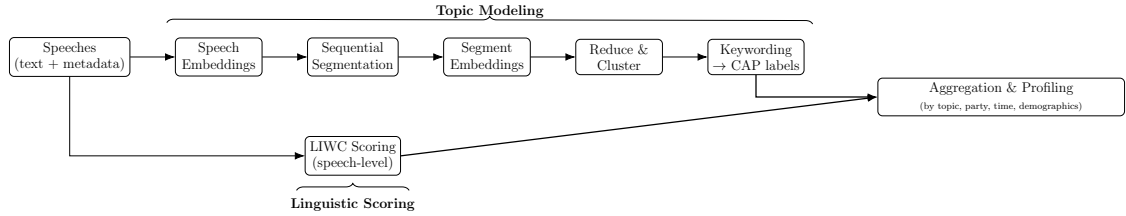


Figure 2: **Segment-first workflow.** Speeches and metadata enter two branches. In the topic branch, speeches are embedded, segmented into agenda episodes, re-embedded at the segment level, reduced, clustered, and mapped to CAP domains. In the style branch, LIWC-22 is computed at the speech level and aggregated by topic, institutional role, and time. The central design choice is to assign topics at the *episode* level rather than the isolated-speech level.

2.4 Linguistic style profiling

LIWC-22 is a dictionary-based tool that maps words into more than 100 interpretable psychological and linguistic categories and returns the proportion of words in a text falling into each [22, 26]. For example, the money category includes terms such as *budget*, *tax*, and *cost*; moral includes *justice*, *right*, and *deserve*; and politic captures the language of political conflict and power. Beyond individual content categories, LIWC-22 provides four summary variables: *Analytic* (formal, logical reasoning), *Clout* (confident, expert-sounding language), *Authentic* (personal, unguarded expression), and *Tone* (overall emotional positivity). The tool has been validated extensively and used across thousands of studies of political and institutional language [22, 26]. We focus on the four summary variables plus money, politic, power, moral, insight, and cogproc as a compact, substantively interpretable set.

Scores are expressed on the LIWC-22 z-scale relative to the Test Kitchen benchmark, a large reference corpus drawn from diverse everyday sources including blogs, emails, and social media. This matters for interpretation: parliamentary language already differs strongly from everyday language, and topic-, role-, and time-specific differences should be read as variation *within* an already distinctive institutional register.

Because LIWC-22 operates on English text, Austrian and Croatian speeches are analyzed through machine translation. This is the approach recommended by the tool’s developers: Boyd et al. [26] explicitly advise translating non-English text to English rather than building native-language dictionaries, on the grounds that the English dictionary has been validated extensively across thousands of studies while translated dictionaries introduce their own reliability costs. We follow that recommendation and provide additional corroboration through cross-lingual embedding validation: for each speech, we embed both the native text and its English translation and compute cosine similarity between the two vectors. Mean similarity is high in both corpora (0.864 for Austria; 0.839 for Croatia), with the mass of the distribution concentrated above 0.8 and somewhat greater variance for very short speeches. Similarity also increases mildly with text length, which is expected because longer texts provide more lexical evidence. Country-wise z-score normalization further reduces any systematic register differences introduced by translation.

2.5 Evaluation philosophy

Our target is episode coherence, while the available benchmarks are speech-level. These are related but not identical tasks. A short speech can look ambiguous in isolation and still belong clearly to an ongoing episode when read in context – the second and third rows of Table 1 illustrate this directly.

A critical point about the human ceiling deserves emphasis. Expert inter-annotator agreement on the available test sets, measured with Krippendorff’s α , is $\alpha \approx 0.60$ – 0.68 [15]. However, those annotators were also reading each speech in isolation, working under exactly the same disadvantage our method is designed to fix. Had they been shown each full sitting in sequential order and asked to label coherent debate chunks rather than individual turns, their agreement would almost certainly have been substantially higher, because context removes a large portion of the ambiguity that drives disagreement. The published Krippendorff’s α should therefore be understood as a ceiling on the isolated-speech task specifically, not as a ceiling on episode-level labelling. Any system evaluated against these benchmarks – ours or others – should be interpreted with that constraint in mind. A fair evaluation of an episode-level method requires episode-level gold labels; building those is the most important methodological next step for this line of work.

We therefore use a mixed evaluation strategy. Macro-F1 against existing labels provides a useful anchor against prior work, while the main analytical question is whether the

method recovers coherent agenda structure that supports interpretable downstream analysis. Against PARLACAP labels and balanced human test sets, the pipeline achieves moderate speech-level agreement (Table 3).

Table 3: Topic classification performance. Human tests are label-balanced and length-filtered. Full-corpus scores use PARLACAP labels. Human inter-annotator agreement (Krippendorff’s α): $\alpha \approx 0.60\text{--}0.68$ [15].

Country	vs. PARLACAP (full corpora)		vs. human tests
	Macro-F1	Micro-F1	Macro-F1
Austria (AT)	0.55	0.57	—
Great Britain (GB)	0.47	0.51	0.43
Croatia (HR)	0.52	0.56	0.47

The more revealing question is where disagreements occur. Figure 3 shows that errors cluster in semantically adjacent domains rather than random ones: Public Lands with Environment, Labour with Social Welfare, International Affairs with Defence. Even when the system misses a boundary, it tends to stay in the correct policy neighborhood; the kind of mistake a careful human annotator reading in isolation would also make, and one that reflects genuine overlap in legislative debate rather than arbitrary failure.

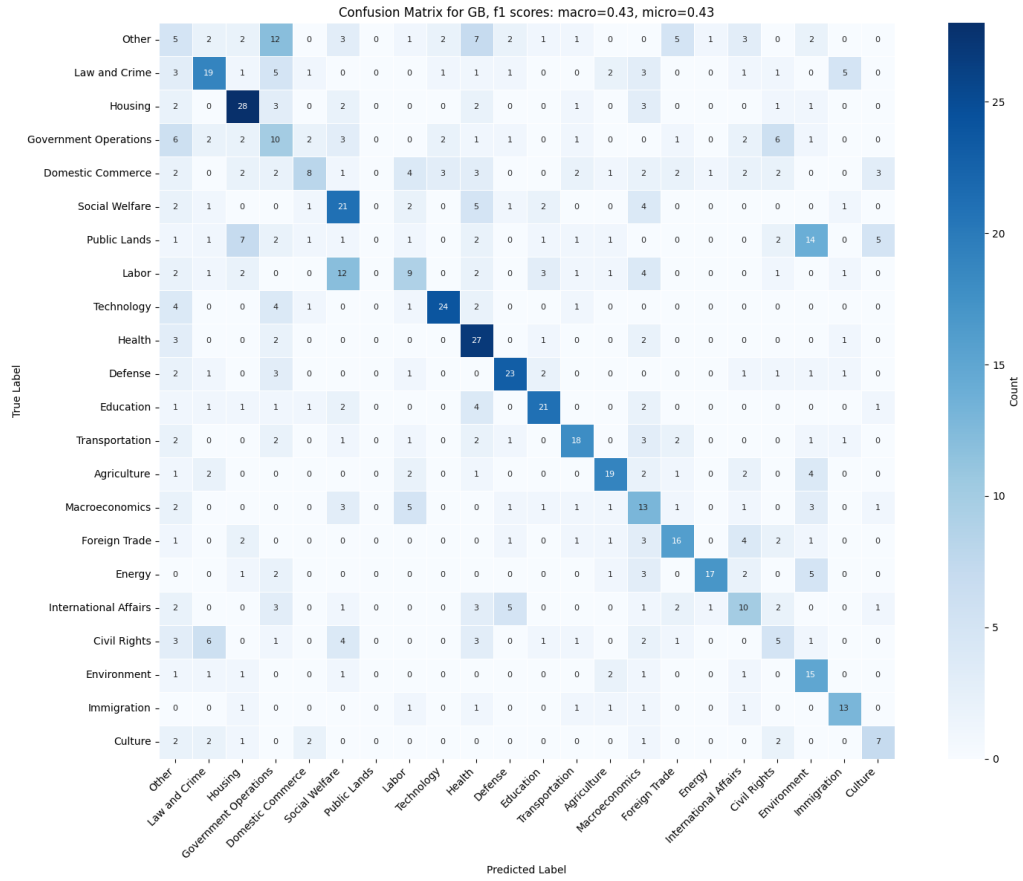


Figure 3: **Confusion structure (Great Britain, human test set).** The diagonal dominates. Off-diagonal mass concentrates in semantically adjacent categories rather than unrelated ones, indicating genuine boundary ambiguity rather than arbitrary misclassification.

We therefore complement benchmark scores with substantive checks: if the topic assignments are meaningful, major policy domains should show strong, interpretable linguistic profiles.

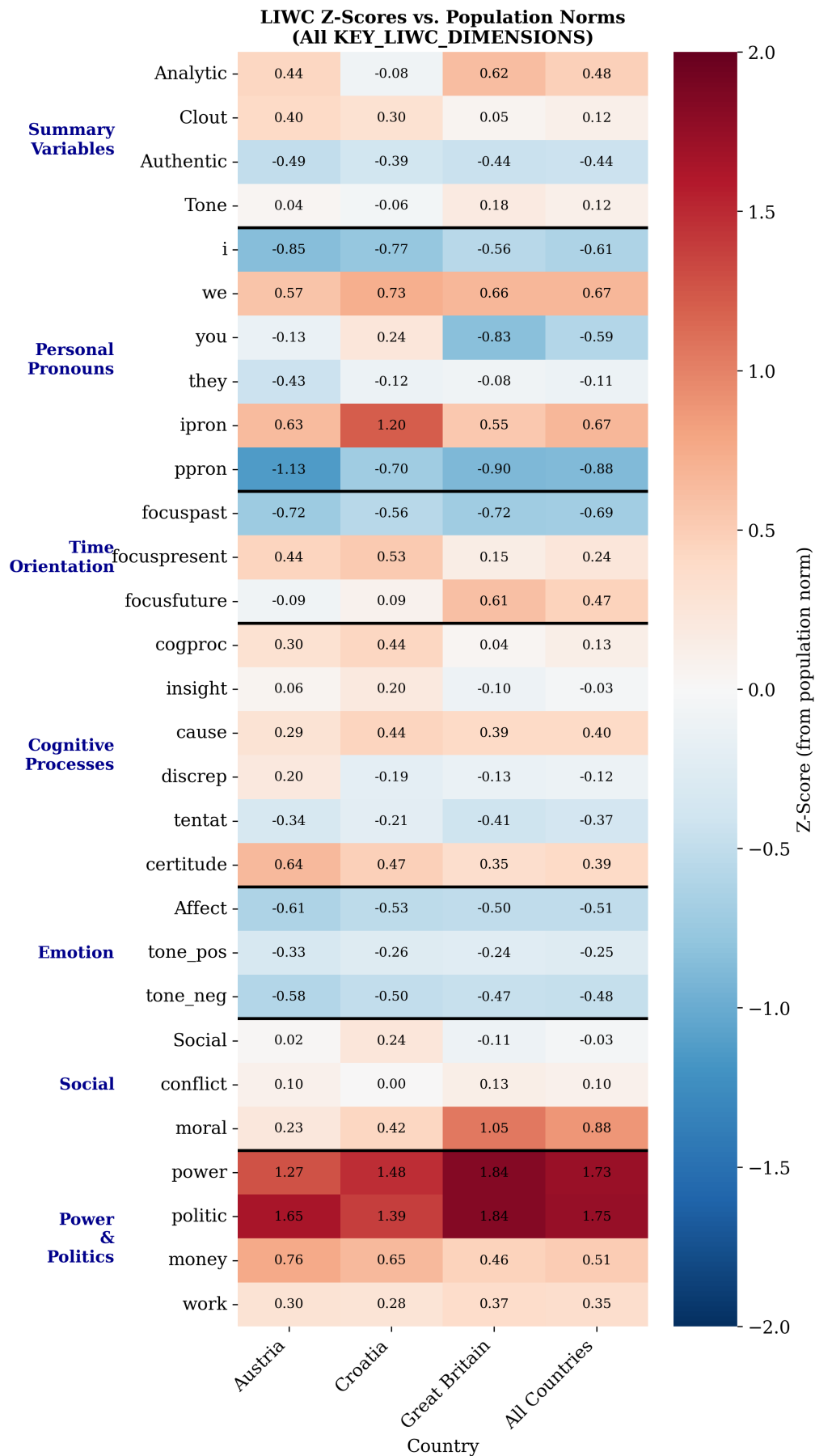


Figure 4: **Parliamentary language versus population norms.** Country-wise LIWC-22 z-scores for all parliamentary speech relative to the Test Kitchen benchmark. Positive values indicate over-representation relative to everyday language; negative values indicate under-use.

3 Results

3.1 Topic styles are distinctive

Before comparing topics, we establish the baseline. Figure 4 shows parliamentary language in each country relative to general-population norms. Across all three corpora, parliamentary speech is elevated on *politic* and *power* and reduced on first-person singular use. This is the institutional baseline against which all subsequent differences should be interpreted.

With that baseline in place, topic differences are clear. Macroeconomics is the strongest and most stable case. Across all three countries, money vocabulary is about 1.4 standard deviations higher than in the rest of the agenda – a large, consistent signal and the primary check for the pipeline. The topic also shows a smaller increase in *Clout* (roughly +0.06 to +0.11), while *moral*, *insight*, and *cogproc* are lower, and *power* is about 0.30 to 0.45 standard deviations lower. The overall profile is concrete, transactional, and authoritative, but less moralized and less overtly adversarial than the parliamentary average.

Health shows the clearest contrast. Relative to other topics, it is lower on *politic*, *power*, and *money*, and slightly higher on *Tone*: health debate is still political, but less conflict-coded and more service-oriented than the parliamentary average. The same logic extends beyond these two focal cases – defense and international affairs are elevated on *politic* and *power*, whereas welfare-facing topics tend to be more social and affective – and each topic has a recognizable rhetorical fingerprint.

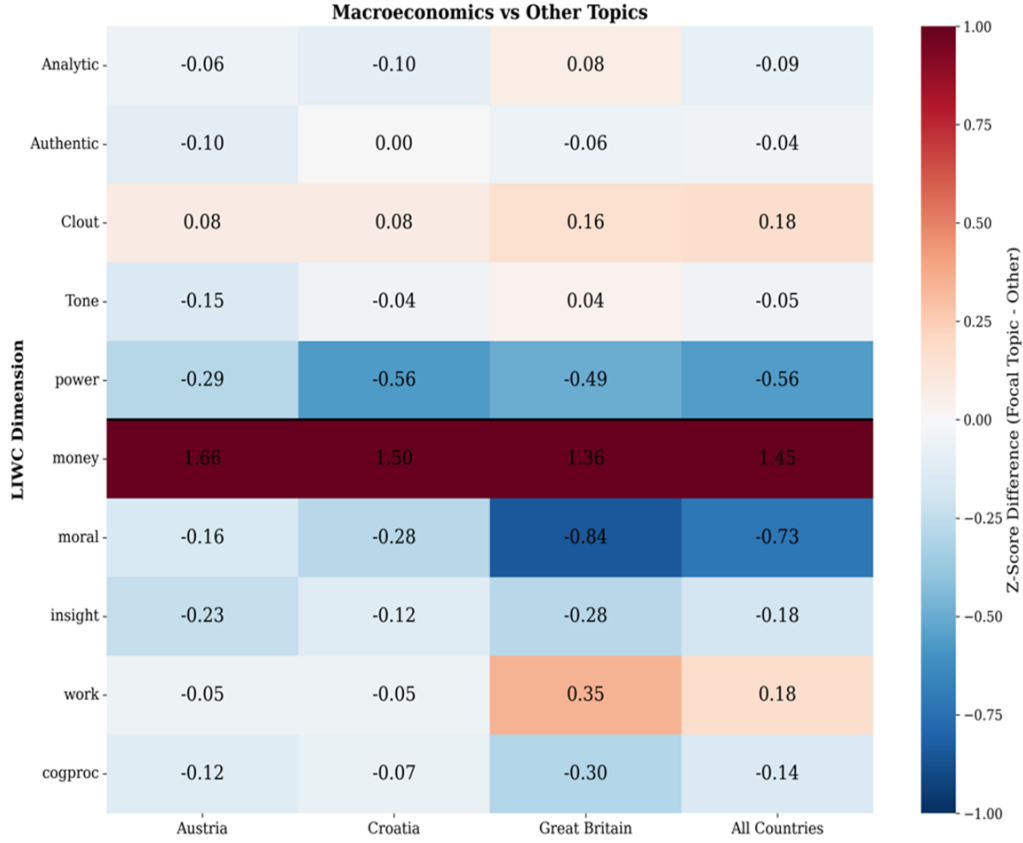


Figure 5: **Macroeconomics versus all other topics.** Cells show Δz values (Macroeconomics minus all other topics) for Austria, Croatia, Great Britain, and the pooled sample. The dominant pattern is the large and highly stable elevation of money language, accompanied by a smaller increase in *Clout* and reductions in *power*, *moral*, *insight*, and *cogproc*.

These topic fingerprints are the first major validation result: the episode labels are not merely internally coherent but substantively useful.

3.2 Institutional role matters more than demographics or ideology

The clearest stylistic contrast in the data is institutional role. Coalition speakers sound more positive and collective; opposition speakers sound more adversarial and explicitly political. This pattern is stable across Austria, Croatia, and Great Britain, and it is larger than the contrasts associated with gender, age, or ideology.

Coalition speech shows higher *Tone* and more we pronouns, while *politic* and *power* are lower. In Great Britain, the coalition–opposition gap in *politic* approaches 0.75 standard deviations. The direction fits the institutional logic: governments project competence and coordination, oppositions project scrutiny and challenge [8, 9, 11].

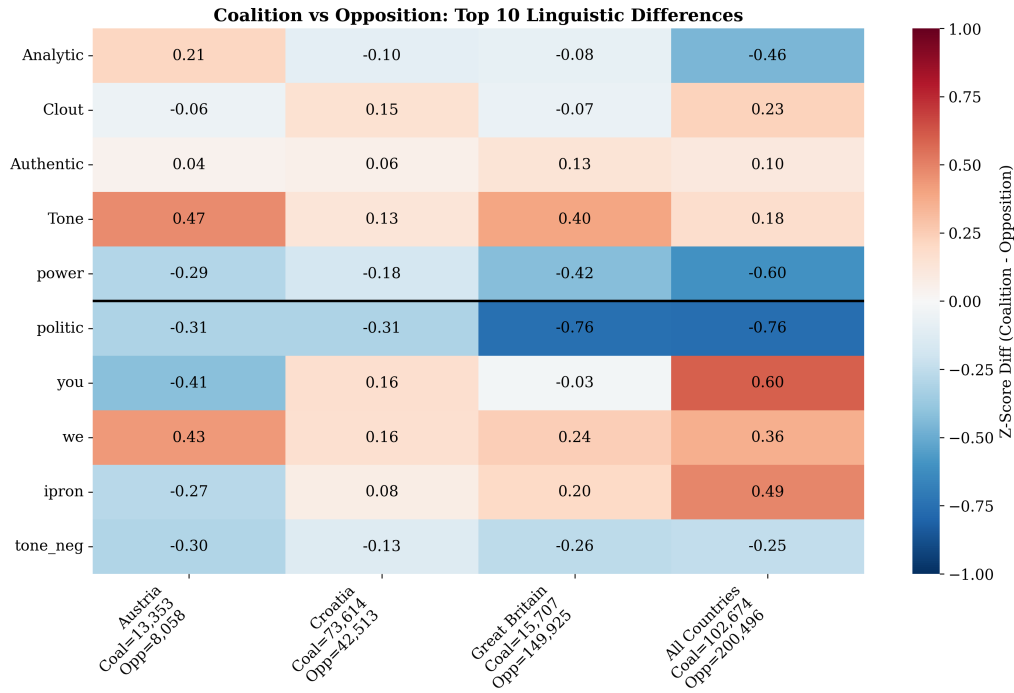


Figure 6: **Coalition versus opposition linguistic differences.** Cells show Δz values (Coalition minus Opposition) for Austria, Croatia, Great Britain, and the pooled sample. Coalition language is more positive (*Tone*) and more collective (*we*); opposition speech is more overtly political and adversarial (*politic*, *power*).

The Austrian party trajectories make the mechanism especially visible. When parties move from opposition into government, tone rises; when they leave government, it falls. This pattern is evident for the Social Democratic Party (SPÖ) and the right-wing Freedom Party (FPÖ), both of which alternate between government and opposition over the sample period. By contrast, the conservative Austrian People's Party (ÖVP), which remains in government throughout, shows no comparable regime change. The simplest reading is that role matters more than party identity.

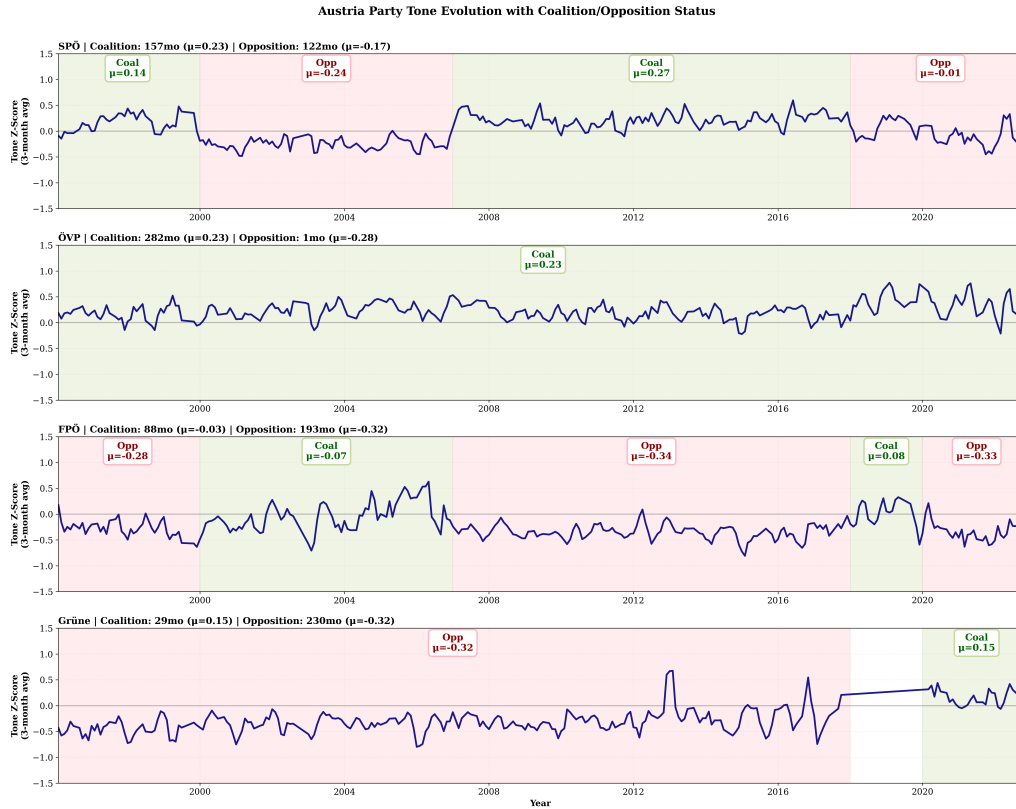


Figure 7: Austrian party tone tracks institutional position. Three-month moving averages of LIWC-22 *Tone* by party. Background shading marks each party’s own periods in government and opposition. Where parties switch roles, tone shifts with the role change rather than with calendar time alone.

Gender differences exist but are smaller. Women show slightly higher *Tone* and somewhat more work- and certainty-related language, while politic and money are marginally lower. Age differences are somewhat clearer: older MPs use more institutional and outward-facing language, especially power and politic, while younger MPs use more first-person singular language and somewhat more *Authentic* and certitude-related expression, consistent with a less established position within the chamber’s institutional hierarchy.

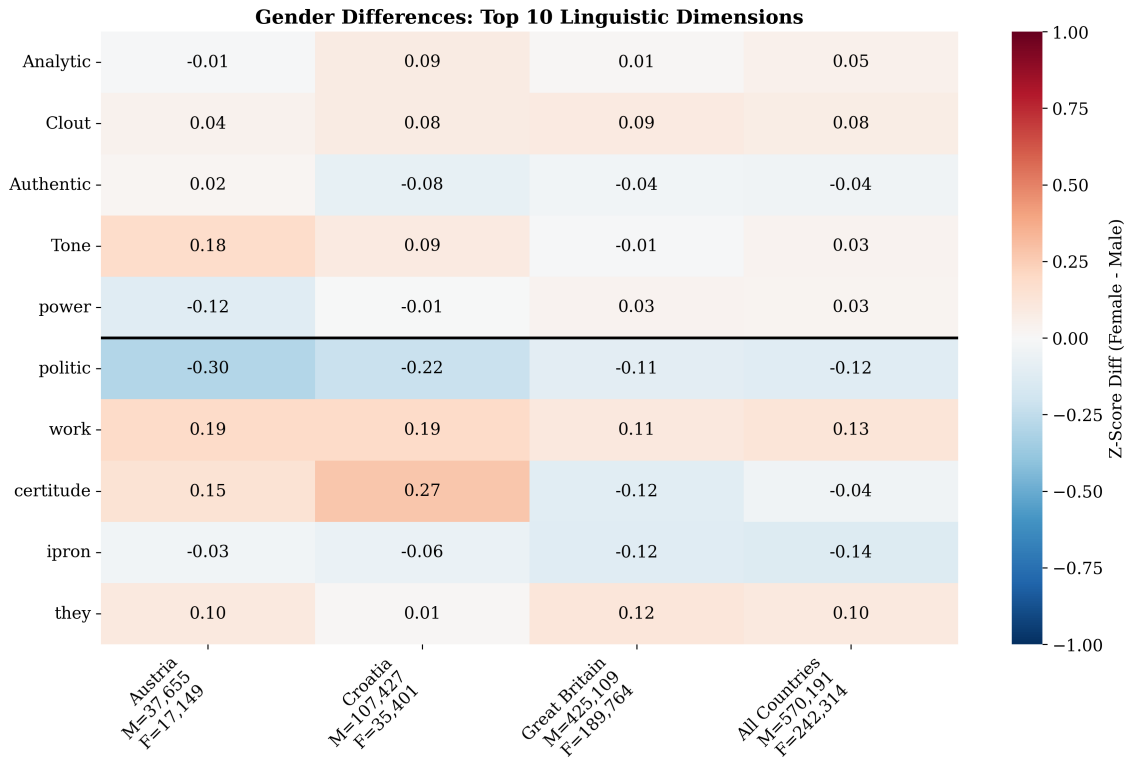


Figure 8: **Gender-related linguistic differences.** Cells show Δz values (women minus men) for Austria, Croatia, Great Britain, and the pooled sample. The differences are modest: women score slightly higher on *Tone* and some work- and certainty-related categories, while *politic* and *money* are somewhat lower.

Perhaps the most striking finding concerns ideology. Across the full left–right spectrum, parliamentary language remains surprisingly stable – far more stable than the coalition–opposition divide, and more stable than prior work on ideological style differences would suggest [23, 24]. The dominant contrast is not left versus right but *parliament versus everyday language*. This implies that the institution itself – its procedural norms, turn-taking rules, and role obligations – imposes a common rhetorical register that cuts across ideological lines. Ideology matters for *what* politicians argue, but the *how* of parliamentary speech appears to be shaped by institutional position far more than by where a speaker sits on the left–right spectrum.

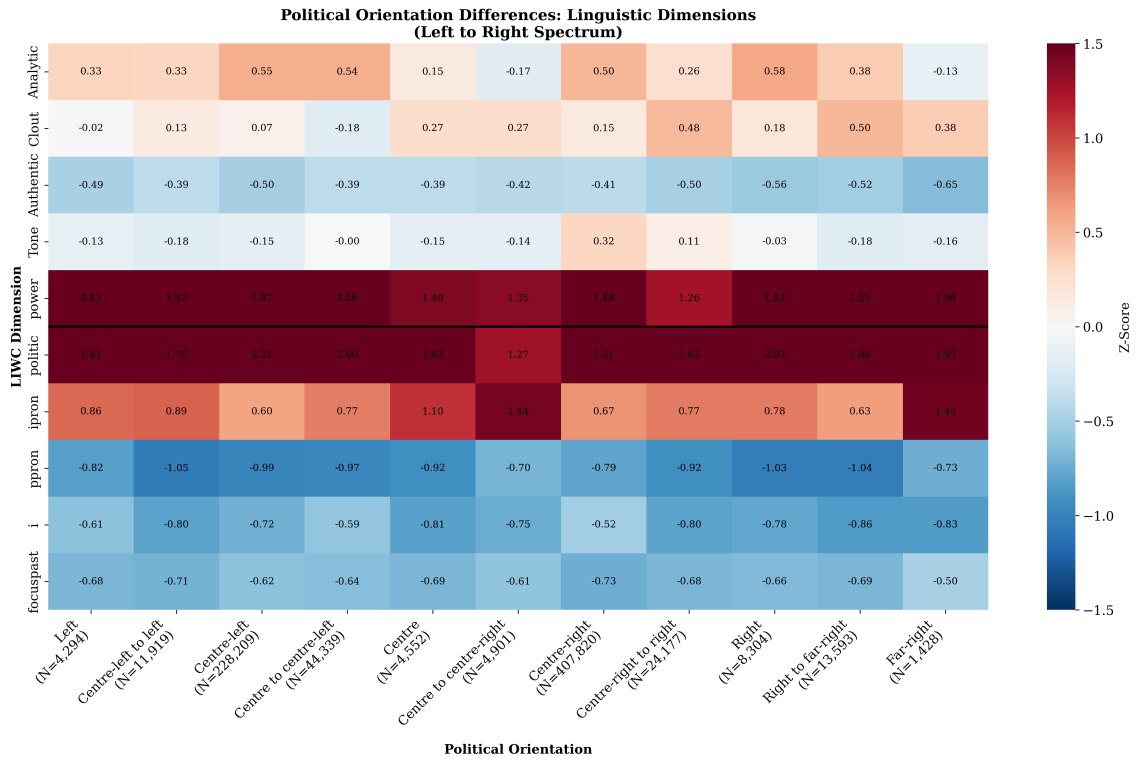


Figure 9: **Linguistic style across the ideological spectrum.** Country-wise LIWC-22 z-scores by left–right party position. Compared with the much larger coalition–opposition contrast, ideological differences are modest and relatively stable, reinforcing the interpretation that parliamentary rhetoric is shaped more by institutional role than by ideological placement alone.

3.3 External shocks reallocate attention and rhetoric – temporarily

Major shocks affect both what parliament discusses and how it speaks, but those changes mostly revert.

The clearest topic-level example is COVID-19. Around 2020, Health rises sharply in all three parliaments while Macroeconomics falls. Austria also shows recurring autumn peaks in Macroeconomics, reflecting the annual budget cycle; Croatia shows a more prolonged health presence. The Ukraine war in 2022 creates a parallel shift in International Affairs and Defence. In each case, the pattern is best described as temporary reallocation rather than permanent restructuring.

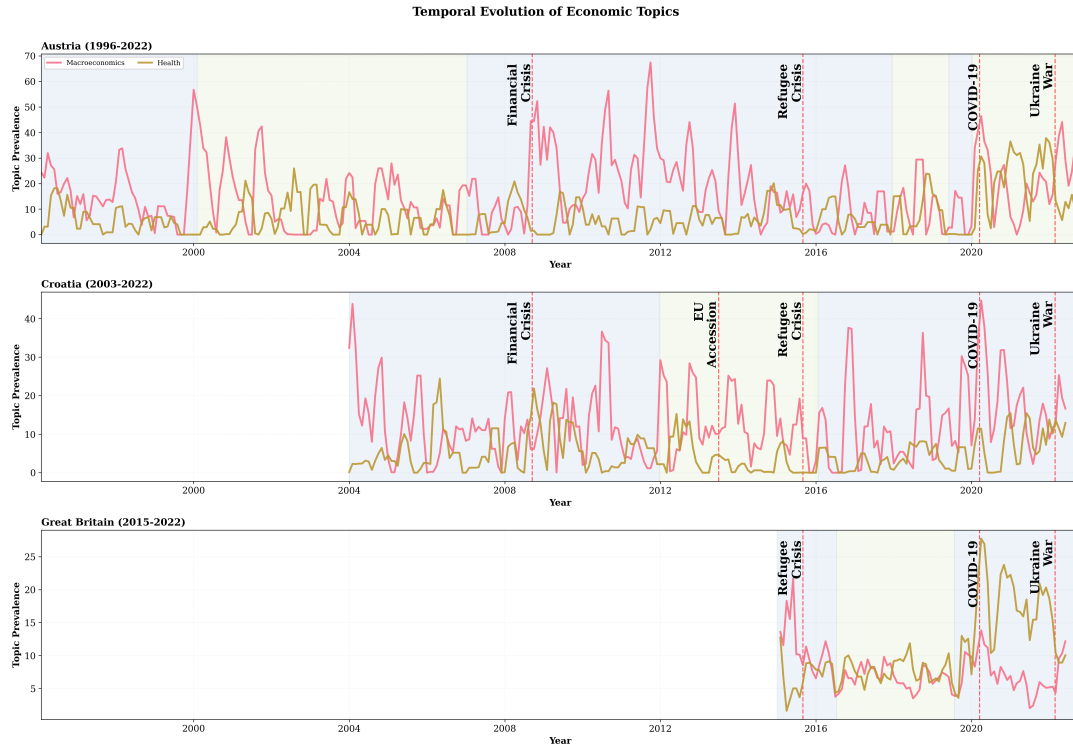


Figure 10: **Agenda dynamics: crises reallocate attention.** Three-month moving averages of topic prevalence for Macroeconomics and Health in Austria, Croatia, and Great Britain. The COVID-19 substitution is visible in all three countries, as is the later return toward baseline. Recurring autumn Macroeconomics peaks reflect the annual budget cycle.

Rhetoric moves in parallel. Politically charged language rises during the 2015 migration crisis, COVID-19 in 2020, and the Ukraine war in 2022, but does not stay permanently elevated. *politic* tends to spike and revert relatively quickly; *moral* vocabulary is more stable and settles back more slowly – a difference in reversion speed that is itself informative, suggesting the institution processes different kinds of political pressure at different rates. Cognitive and explanatory language also rises during moments of intense policy justification, especially in the early pandemic period. The pattern matches the broader argument of the paper: crises temporarily intensify both agenda focus and rhetorical contestation, but the chamber largely returns to its institutional baseline after the acute phase passes. That reversion is most clearly visible at the episode level; speech-by-speech analysis would be too noisy to detect it reliably.

4 Discussion and conclusion

The choice of analytical unit is not just a technical convenience. If meaning in parliament is produced through structured, sequential exchanges governed by procedural rules and role obligations, then breaking that exchange into isolated turns strips away precisely the context that makes parliamentary language political in the first place. The results reported

here support that argument from three angles, and each angle has implications beyond the immediate methodological contribution.

First, the recovered episode labels align with strong substantive expectations. Macroeconomics is consistently money-heavy and less moralized across all three countries; health debate is less conflict-coded and slightly more positive; defense and international affairs are elevated on political and power language; welfare-facing topics are more social and affective. Each major policy domain has a recognizable rhetorical fingerprint, and the consistency of those fingerprints across Austria, Croatia, and Great Britain – three different languages, institutional cultures, and political histories – is the primary substantive validation of the pipeline.

Second, the largest stylistic divide is institutional rather than demographic or ideological. Coalition and opposition speakers differ more than men and women, younger and older MPs, or parties on the left and right of the spectrum. The weakness of ideology as a predictor of rhetorical style is the most striking of these findings. Prior work has documented ideological style differences in legislative speech [23, 24], and the expectation that left and right parties would sound systematically different is not unreasonable. That the coalition–opposition divide dwarfs the ideological one suggests something stronger: the institution itself—its norms, rituals, and differentiated role obligations—standardizes how politicians speak in ways that cross-cut ideological identity. This interpretation is broadly consistent with prior work showing that institutional context shapes legislative speech, although most existing studies emphasize ideological differences rather than role-based ones. Whether the pattern observed here reflects institutional socialization over time, selection into parliamentary careers, or strategic adaptation to chamber norms is a question the present data cannot fully resolve. However, its stability across three countries suggests that it reflects a genuine empirical regularity rather than a methodological artifact, and it invites more direct, targeted investigation in future work.

Third, crises temporarily reorganize both agenda attention and rhetoric, but the chamber largely returns to its institutional baseline after the acute phase passes. The health surge during COVID-19, the security shift after the Ukraine war, and the rise and later decline of overtly political rhetoric all become easier to read once debate is organized into coherent agenda chunks. The difference in reversion speed between *politic* and *moral* vocabulary is itself informative: the former spikes and reverts quickly, while the latter settles back more slowly. This suggests the institution processes different kinds of political pressure at different rates, and that crisis rhetoric is not a single uniform phenomenon.

Two limitations deserve mention before turning to extensions. The available evaluation data are speech-level rather than episode-level, so validation necessarily combines benchmark metrics with substantive interpretive checks. Building episode-level gold labels – ideally through annotation protocols that show coders full sequential debate chunks rather than isolated turns – is the most important methodological next step for this line of work.

The broader relevance of results also remains to be tested: the three-country sample is deliberately chosen for variation in language, institutional design, and political context, but it is still a sample.

Those limitations point directly toward the most productive extensions of the framework. The most immediate is scale: the pipeline can be applied to the full set of 29 European parliaments available in PARLAMINT, enabling a substantially wider comparative analysis of agenda structure and rhetorical style across institutional contexts that vary in terms of procedural rules, dynamics and style of plenary debates or in the role of plenary debates vs committee debates. A second extension could be in a more predictive direction: if particular combinations of topic, tone, moralization, conflict language, and certainty are systematically associated with legislative success or failure, those patterns could be modeled explicitly rather than only observed after the fact.

The broader methodological contribution is not limited to parliamentary data. The underlying logic of the episode-level approach applies wherever topical structure emerges through interaction rather than being pre-annotated in advance: committee hearings, council meetings, court transcripts, broadcast debates, customer-support conversations, online discussion threads, and other forms of multi-party institutional dialogue. In all of these settings, topical boundaries are not given – they are produced, negotiated, and sometimes blurred in the course of the exchange itself. The combination of procedural cues and semantic drift used here offers a practical way to recover those boundaries and assign coherent discussion-level labels with minimal human effort. In that sense, the pipeline is not only a relevant contribution to a parliamentary labelling problem but a more general approach to recovering topical structure in sequential conversational data – and the substantive findings reported here suggest that recovering that structure is worth the effort.

Statements and Declarations

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Data Availability

The PARLAMINT corpora analyzed in this study are publicly available from the Slovenian Language Resource Repository CLARIN.SI (<https://www.clarin.si/repository/xmlui/>), specifically the ParlaMint-en.ana 5.0 deposit (<http://hdl.handle.net/11356/2006>). The

PARLACAP labels and human-labelled test sets used for evaluation are available from Kuzman Pungertšek et al. [15]. LIWC-22 is a proprietary tool available from <https://www.liwc.app/>.

All code used to produce the segmentation, clustering, CAP mapping, and LIWC aggregation results in this paper is publicly available at <https://github.com/pavlesav/master-thesis>.

Author Contributions

Conceptualization: P. Savković, D. Helic

Methodology: P. Savković

Software: P. Savković

Formal analysis and investigation: P. Savković

Writing – original draft preparation: P. Savković

Writing – review and editing: P. Savković, S. Thalmann, A. Spök, D. Helic

Supervision: D. Helic, S. Thalmann, A. Spök

Use of Generative AI

Generative AI tools (GPT and Claude) were used during manuscript preparation for language editing, formatting assistance, and structural revision. All scientific content, results, interpretations, and conclusions are the authors' own. The cluster keyword-to-CAP mapping step described in Section 2 used a large language model as part of the research methodology; the full prompt template is reproduced in Appendix A, and all model inputs and outputs were logged for auditability.

A Prompt for LLM

The following prompt template was used in the cluster-labelling step. Placeholder fields for country, source language, keywords, and CAP category descriptions were filled automatically for each cluster.

Analyze these parliamentary keywords and provide TWO outputs IN ENGLISH:

Country: [country] Parliament

Source Language: [language]

Keywords: [comma-separated cluster keywords]

TASK 1 – Topic Name: Create a short, descriptive name IN ENGLISH (2–4 words) that captures what this topic is about in the parliamentary debate.

TASK 2 – CAP Classification: Classify into ONE of these policy categories:

[full list of CAP categories and descriptions inserted here]

Instructions: always respond in English; provide a specific and descriptive topic name; choose the most specific policy category; use Other for procedural or non-policy content.

Format your response exactly as:

TOPIC: [your English topic name]

CATEGORY: [exact category name from list]

References

- [1] Cornelia Ilie, ed. *European Parliaments under Scrutiny: Discourse Strategies and Interaction Practices*. Amsterdam: John Benjamins, 2010.
- [2] Paul Bayley, ed. *Cross-Cultural Perspectives on Parliamentary Discourse*. Amsterdam: John Benjamins, 2004.
- [3] Teun A. van Dijk. “Text and Context of Parliamentary Debates”. In: *Cross-Cultural Perspectives on Parliamentary Discourse*. Ed. by Paul Bayley. Amsterdam: John Benjamins, 2004, pp. 339–372.
- [4] Justin Grimmer. “A Bayesian Hierarchical Topic Model for Political Texts: Measuring Expressed Agendas in Senate Press Releases”. In: *Political Analysis* 18.1 (2010), pp. 1–35.
- [5] Kevin M. Quinn, Burt L. Monroe, Michael Colaresi, Michael H. Crespin, and Dragomir R. Radev. “How to Analyze Political Attention with Minimal Assumptions and Costs”. In: *American Journal of Political Science* 54.1 (2010), pp. 209–228.
- [6] Justin Grimmer and Brandon M. Stewart. “Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts”. In: *Political Analysis* 21.3 (2013), pp. 267–297.
- [7] Norman Fairclough. *Language and Power*. London: Longman, 1989.
- [8] Sven-Oliver Proksch and Jonathan B. Slapin. “Position Taking in European Parliament Speeches”. In: *British Journal of Political Science* 40.3 (2010), pp. 587–611.
- [9] Benjamin E. Lauderdale and Alexander Herzog. “Measuring Political Positions from Legislative Speech”. In: *Political Analysis* 24.3 (2016), pp. 374–394.
- [10] Gavin Abercrombie and Riza Batista-Navarro. ““Aye” or “No”? Speech-Level Sentiment Analysis of Hansard UK Parliamentary Debate Transcripts”. In: *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*. Miyazaki, Japan: European Language Resources Association, 2018.

- [11] Ludovic Rheault, Kaspar Beelen, Christopher Cochrane, and Graeme Hirst. “Measuring Emotion in Parliamentary Debates with Automated Text Analysis”. In: *PLOS ONE* 11.12 (2016), e0168843.
- [12] Gavin Abercrombie and Riza Batista-Navarro. “Sentiment and Position-Taking Analysis of Parliamentary Debates: A Systematic Literature Review”. In: *Journal of Computational Social Science* 3.1 (2020), pp. 245–270.
- [13] Tomaž Erjavec et al. “ParlaMint II: Advancing Comparable Parliamentary Corpora Across Europe”. In: *Language Resources and Evaluation* 59.3 (2025), pp. 2071–2102. DOI: 10.1007/s10579-024-09798-w.
- [14] Taja Kuzman Pungeršek et al. *Linguistically Annotated Multilingual Comparable Corpora of Parliamentary Debates in English ParlaMint-en.ana 5.0*. Slovenian Language Resource Repository CLARIN.SI. Data set. 2025. URL: <http://hdl.handle.net/11356/2006>.
- [15] Taja Kuzman Pungeršek, Peter Rupnik, Daniela Širinić, and Nikola Ljubešić. *Supercharging Agenda Setting Research: The ParlaCAP Dataset of 28 European Parliaments and a Scalable Multilingual LLM-Based Classification*. 2026. arXiv: 2602.16516.
- [16] Frank R. Baumgartner, Bryan D. Jones, and John Wilkerson. *Policy Agendas Codebook*. 2019.
- [17] Marti A. Hearst. “TextTiling: Segmenting Text into Multi-Paragraph Subtopic Passages”. In: *Computational Linguistics* 23.1 (1997), pp. 33–64.
- [18] Freddy Y. Y. Choi. “Advances in Domain Independent Linear Text Segmentation”. In: *Proceedings of NAACL 2000*. 2000, pp. 26–33.
- [19] Martin Riedl and Chris Biemann. “TopicTiling: A Text Segmentation Algorithm Based on LDA”. In: *Proceedings of the ACL 2012 Student Research Workshop*. Association for Computational Linguistics, 2012, pp. 37–42.
- [20] Jacob Eisenstein and Regina Barzilay. “Bayesian Unsupervised Topic Segmentation”. In: *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing (EMNLP 2008)*. 2008, pp. 334–343.
- [21] Omri Koshorek, Adir Cohen, Noam Mor, Michael Rotman, and Jonathan Berant. *Text Segmentation as a Supervised Learning Task*. 2018. arXiv: 1803.09337.
- [22] Yla R. Tausczik and James W. Pennebaker. “The Psychological Meaning of Words: LIWC and Computerized Text Analysis Methods”. In: *Journal of Language and Social Psychology* 29.1 (2010), pp. 24–54.

- [23] Jacob Jensen, Ethan Kaplan, Suresh Naidu, and Laurence Wilse-Samson. “Political Polarization and the Dynamics of Political Language: Evidence from 130 Years of Partisan Speech”. In: *Brookings Papers on Economic Activity* 43.2 (2012), pp. 1–81.
- [24] Martijn Schoonvelde, Anna Brosius, Gijs Schumacher, and Bert N. Bakker. “Liberals Lecture, Conservatives Communicate: Analyzing Complexity and Ideology in 381,609 Political Speeches”. In: *PLOS ONE* 14.2 (2019), e0208450.
- [25] Camilla Del Gennaro, Marco Zanetti, Marco Guerini, and Piercesare Nobili. “Morality in Political Communication: A Cross-Country, Cross-Party Analysis”. In: *Political Communication* 38.5 (2021), pp. 543–562.
- [26] Ryan L. Boyd, Ashish Ashokkumar, Sarah Seraj, and James W. Pennebaker. *The Development and Psychometric Properties of LIWC-22*. Technical manual. University of Texas at Austin. Austin, TX, 2022.
- [27] Kayla N. Jordan, Joan Sterling, James W. Pennebaker, and Ryan L. Boyd. “Examining Long-Term Trends in Politics and Culture through Language of Political Leaders and Cultural Institutions”. In: *Proceedings of the National Academy of Sciences* 116.9 (2019), pp. 3476–3481.
- [28] Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. “BGE M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation”. In: *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, 2024.
- [29] Leland McInnes, John Healy, and James Melville. *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*. 2018. arXiv: 1802.03426.
- [30] Douglas A. Reynolds. “Gaussian Mixture Models”. In: *Encyclopedia of Biometrics*. Ed. by Stan Z. Li and Anil K. Jain. New York: Springer, 2009, pp. 659–663.
- [31] Maarten Grootendorst. *BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure*. 2022. arXiv: 2203.05794.